

COCHABAMBA, Bolivia

WMO: 852230

Lat: 17.421S

Lon: 66.177W

Elev: 2548

StdP: 74.23

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	2.1	3.8	-9.0	2.4	22.6	-6.2	3.1	19.4	9.1	18.6	7.8	19.8	0.1	300	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	14.6	30.1	14.6	29.2	14.4	28.2	14.2	17.2	25.9	16.7	25.3	16.2	24.7	3.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	14.5	19.7	14.1	13.8	18.4	13.9	13.6	18.1	60.0	26.0	58.2	25.6	56.3	24.9	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.4	5.7	-0.7	32.5	1.3	1.0	-1.6	33.2	-2.4	33.7	-3.1	34.3	-4.1	35.0
			-2.5	19.3	1.0	1.8	-3.2	20.6	-3.8	21.6	-4.4	22.6	-5.1	24.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	19.1	19.0	19.0	18.3	16.2	14.5	14.6	16.1	18.1	19.7	20.4	20.0		
	DBStd	2.55	1.94	1.82	1.80	1.48	1.57	1.48	1.36	1.39	1.46	1.44	1.69	1.91		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	463	13	13	14	18	68	114	116	69	21	5	4	7		
	CDD10.0	2888	283	251	279	249	193	136	142	190	242	301	312	310		
	CDD18.3	310	38	31	34	17	2	0	0	1	14	47	67	59		
	CDH23.3	4447	277	234	324	389	298	178	179	329	443	619	674	501		
(13)	CDH26.7	778	45	34	53	58	28	10	9	43	77	138	172	111		
(14)	Wind	WSAvg	1.6	1.7	1.5	1.4	1.3	1.1	0.9	1.1	1.5	2.1	2.2	2.2		
(15) Precipitation	PrecAvg	472	121	92	69	25	5	1	1	4	8	15	40	95		
	PrecMax	706	227	270	164	76	31	9	10	32	36	50	113	199		
	PrecMin	268	29	15	11	0	0	0	0	0	0	0	4	35		
	PrecStd	103	55	50	41	20	7	2	3	7	9	13	25	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	29.5	30.0	29.5	28.8	27.9	27.9	29.2	30.1	31.0	31.3	31.0		
		MCWB	16.4	16.3	16.0	14.3	12.6	11.8	11.3	11.9	13.2	13.7	14.4	15.4		
	2%	DB	27.9	27.8	28.1	28.1	27.2	26.8	26.7	27.9	28.8	29.2	30.0	29.1		
		MCWB	15.5	15.9	15.6	14.3	12.5	11.5	11.1	11.8	13.2	14.1	14.6	15.2		
	5%	DB	26.2	26.1	26.8	27.1	26.2	25.3	25.2	26.8	27.2	28.1	28.8	27.9		
		MCWB	15.1	15.5	15.5	14.3	12.3	11.3	11.1	11.8	13.0	14.0	14.6	15.0		
(25) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	24.8	24.8	25.1	25.9	25.1	24.1	24.1	25.2	26.1	27.0	27.2	26.2		
		MCWB	15.1	15.4	15.2	14.0	12.3	11.2	11.1	11.7	12.9	13.9	14.5	14.8		
	0.4%	WB	17.9	18.1	17.7	17.0	15.2	14.1	13.8	14.7	15.9	16.8	17.4	17.7		
		MCDB	26.5	26.6	26.2	25.3	24.9	23.9	24.5	25.5	26.1	27.1	27.3	26.9		
	2%	WB	16.9	17.0	16.8	16.0	14.2	13.1	12.9	13.7	14.9	15.9	16.6	16.7		
		MCDB	24.7	24.7	25.3	25.0	24.5	24.1	23.6	24.8	25.8	25.9	26.4	25.9		
(30) Mean Daily Temperature Range	5%	WB	16.1	16.3	16.1	15.3	13.6	12.4	12.2	12.9	14.1	15.1	15.9	16.1		
		MCDB	23.7	23.6	24.5	24.4	23.9	23.3	23.0	24.4	25.0	25.5	25.5	25.1		
	10%	WB	15.5	15.8	15.5	14.7	13.0	11.7	11.5	12.3	13.4	14.5	15.1	15.5		
		MCDB	22.5	22.7	23.3	23.8	23.0	22.3	22.0	23.4	24.1	24.6	24.7	23.9		
	5% DB	MDBR	11.3	11.5	12.7	16.2	19.6	21.0	20.7	19.9	17.4	15.7	14.6	13.0		
		MCDBR	14.0	14.2	15.4	18.3	20.9	22.5	22.5	22.0	19.4	17.7	17.2	15.6		
(36) Clear-Sky Solar Irradiance	5% WB	MCWBR	4.6	4.9	5.2	6.5	8.3	9.6	9.6	8.7	7.3	6.0	5.6	5.1		
		MCDBR	12.1	12.3	13.6	15.9	18.7	20.8	20.7	19.9	17.7	15.7	14.7	13.7		
	taub	taub	0.341	0.342	0.329	0.308	0.297	0.292	0.299	0.368	0.454	0.409	0.359	0.356		
		taud	2.492	2.500	2.543	2.589	2.553	2.584	2.523	2.273	2.017	2.206	2.377	2.420		
(42) All-Sky Solar Radiation	Edn at Noon	Edn at Noon	1002	991	982	968	944	932	932	891	849	916	978	987		
		Edh at Noon	117	114	106	94	91	85	93	128	176	152	130	125		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.32	5.21	5.29	5.24	4.98	4.69	4.85	5.33	5.62	5.84	6.12	5.56		
		RadStd	0.39	0.30	0.34	0.31	0.23	0.23	0.24	0.27	0.41	0.37	0.34	0.28		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.29	+0.71	-3.10	N/A	N/A	N/A	N/A	-82	+109	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page